Embedded System

Automatic Vehicle Speed Controller System

Project Proposal

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ABSTRACT:

Automatic Vehicle Speed Controller in Traffic Using RF Signal and 8051 Microcontroller. This project can be used to control the speed of the vehicle by reducing the acceleration of the vehicle near traffic signals. The light on the traffic signal will determine the speed of the vehicle. The microcontroller will reduce the acceleration in case of yellow light and completely stops the vehicle in case of red light. An RF transmitter is located near the traffic lights and a corresponding RF receiver is placed in the vehicle.

INTRODUCTION:

The project is "Automatic Vehicle Speed Controller System" which is very beneficial to improve Road Safety, the way it automatically controls speed limits of a vehicle installed in it. The system, with the help of speed sensors, micro controllers, and actuators, detects speed values and it regulates speed on further environmental conditions such as road zones or traffic conditions, and thus avoids accidental over-speeding. This ensures compliance with traffic regulations, thus minimizing the risk of accidents and is particularly useful in school zones or along highways or construction sites. It is going to be utterly compatible for modern vehicles and will prove to be an effective, practical, and hassle-free solution to speed management.

OBJECTIVES:

1.Speed Regulation and Safety Standards in Modern Transport Systems Studies.

2.Identify the key components such as sensors, microcontrollers, and actuators, that are necessary for automating speed control.

3.Design a zone speed detector system that autonomously adjusts the speed of the vehicle according to the prescribed laws.

4.Analyze performance of the system for different road and traffic conditions, ensuring reliability and efficiency.

5.Compare the efficiency of the proposed system with conventional speed control systems to draw the benefits in safety and compliance.

PROBLEM STATEMENT:

The increased incidence of road accidents due to over-speeding is an imminent threat to public safety, necessitating effective speed management solutions. Obviously, enforced methods of speed limit enforcement, such as road signs and manually monitoring speed, are less effective in areas where compliance is most necessary, such as school zones, highways, and construction sites. Thus, this problem is being tackled by designing an Automatic Vehicle Speed Controller System that autonomously controls the speed of a vehicle according to predefined limits, which reduces human error, enhances safety, and improves compliance with traffic regulations.

METHODOLOGY:

The methodology for the Automatic Vehicle Speed Controller System includes design of the system architecture, sensor and microcontroller installation, software development, testing and calibrating the system, and finally deploying it in prototype vehicle for evaluation in real-world situations. This system is programmed to process sensor input and execute speed control commands following algorithms based on speed zone data. The next step is testing and calibration by means of controlled environment in order to provide detection of speed limits and adjustment of speed reliably. Finally, it will be deployed in prototype vehicle and evaluated in real-world scenarios.

COMPONENTS USED:

1.PROTEUS

2.MICROCONTROLLER

3.SPEED CONTROLLER

4.GPS MODULE

5.RF TRANSMITTER AND RECEIVER

6.ACTUATORS

7.POWER SUPPLY

8.DISPLAY MODULE

9.KEIL AND ARDUINO IDE

CONCLUSION:

In conclusion, the Automatic Vehicle Speed Controller System is a revolutionary tool that automates speed regulation in line with traffic rules. It uses sensors, microcontrollers, and actuators to detect speed limits and adjust the vehicle's speed, reducing over-speeding risks and enhancing road safety. The prototype shows potential for improving compliance in high-risk areas like school zones and construction sites. Future developments could include real-time traffic data integration and improved adaptability to different road conditions.